

Loan Approval Prediction

A Machine Learning Classification Project Report

Predicting loan approval outcomes using applicant financial data

Dataset: 4,269 loan applications, 13 columns

Task: Supervised binary classification (Approved / Rejected)

Best model: Random Forest | Test accuracy 98.13% | AUC 0.998

1. Executive Summary

This project builds a supervised machine learning system that predicts whether a bank loan application should be **Approved** or **Rejected**, based on an applicant's personal and financial profile. The objective is to reduce manual verification effort, speed up loan decisions, and minimize the risk of approving high-risk applicants.

Seven classification algorithms were trained and compared on a dataset of 4,269 loan applications. The Random Forest classifier delivered the best out-of-the-box performance (98.13% accuracy). The tuned Random Forest achieved an AUC of 0.998, indicating near-perfect separation between approved and rejected applications. The CIBIL (credit) score emerged as by far the most influential predictor of loan approval.

98.13% Best test accuracy (Random Forest)	0.998 AUC score (tuned model)	81.9% CIBIL importance (Random Forest)	4,269 Applications analysed
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2. Business Understanding

2.1 Problem Statement

A bank wants to predict whether a loan application should be approved based on the applicant's personal and financial information. This helps reduce manual effort and speeds up the loan approval process.

2.2 Business Objectives

- Reduce manual verification of loan applications
- Minimize the risk of approving high-risk applicants
- Improve customer experience through faster turnaround

2.3 Machine Learning Problem Framing

- **Type:** Supervised Learning
- **Category:** Binary Classification
- **Target variable:** loan_status (Approved / Rejected)

3. Data Understanding

The dataset consists of 4,269 loan applications with 13 columns describing each applicant's demographic and financial profile.

Feature	Description
loan_id	Unique application identifier
no_of_dependents	Number of dependents the applicant has
education	Graduate / Not Graduate
self_employed	Whether the applicant is self-employed
income_annum	Annual income of the applicant
loan_amount	Loan amount requested
loan_term	Loan repayment term (years)

Feature	Description
cibil_score	Credit score indicating creditworthiness
residential_assets_value	Value of residential assets
commercial_assets_value	Value of commercial assets
luxury_assets_value	Value of luxury assets
bank_asset_value	Bank balance / financial assets
loan_status (Target)	Approved (1) or Rejected (0)

4. Data Cleaning & Preparation

- No duplicate records were found in the dataset.
- No missing values were present across any column.
- Column names were stripped of leading/trailing whitespace.
- Columns were renamed for clarity: loan_term to loan_term_years, bank_asset_value to bank_balance.

5. Exploratory Data Analysis

5.1 Univariate Analysis

- Annual income is broadly spread across low, medium, and high-income applicants.
- Most applicants have a CIBIL score above 700, with a secondary cluster between 500 and 600.
- Most requested loan amounts fall between 1 and 2 crore; amounts above 2.5 crore are rare.
- Graduates and non-graduates are represented in almost equal numbers, with a slight edge to graduates.
- Self-employed and salaried applicants are represented in almost equal proportion.
- Approved loans outnumber rejected loans in the dataset.
- Boxplots of annual income and loan amount show no significant outliers.

5.2 Bivariate Analysis

- Education level shows little to no impact on loan approval outcomes.
- Self-employment status shows little to no impact on loan approval outcomes.
- Approved applicants tend to have a noticeably higher CIBIL score than rejected applicants; a small number of high-CIBIL applicants are still rejected.
- Average annual income is similar between approved and rejected groups, suggesting income alone is not decisive.

5.3 Multivariate Analysis

- A pairplot of income, loan amount, CIBIL score, and bank balance shows that a CIBIL score above ~600 is strongly associated with approval.
- Higher bank balance is associated with larger approved loan amounts.
- Income and bank balance rise together.

5.4 Correlation Analysis

- Annual income and loan amount are highly positively correlated; income has the strongest linear relationship with loan amount.
- Luxury asset value and annual income are highly positively correlated.
- Bank balance and income also show high correlation.

- `loan_id` and `no_of_dependents` contribute little signal and are treated as low-value features. Note that `loan_term_years` shows weak linear correlation with other variables but still ranks highly in model-based feature importance (Section 10), so it is retained.

6. Feature Engineering

6.1 Encoding

Categorical variables (`education`, `self_employed`, `loan_status`) were converted to numeric form using Label Encoding so they could be consumed by the machine learning models.

6.2 Feature Scaling

Numerical features (`income_annum`, `loan_amount`, `cibil_score`, `residential_assets_value`, `commercial_assets_value`, `luxury_assets_value`, `bank_balance`) were standardized using `StandardScaler`, which benefits distance- and gradient-based algorithms such as KNN, SVM, and Logistic Regression.

6.3 Train-Test Split

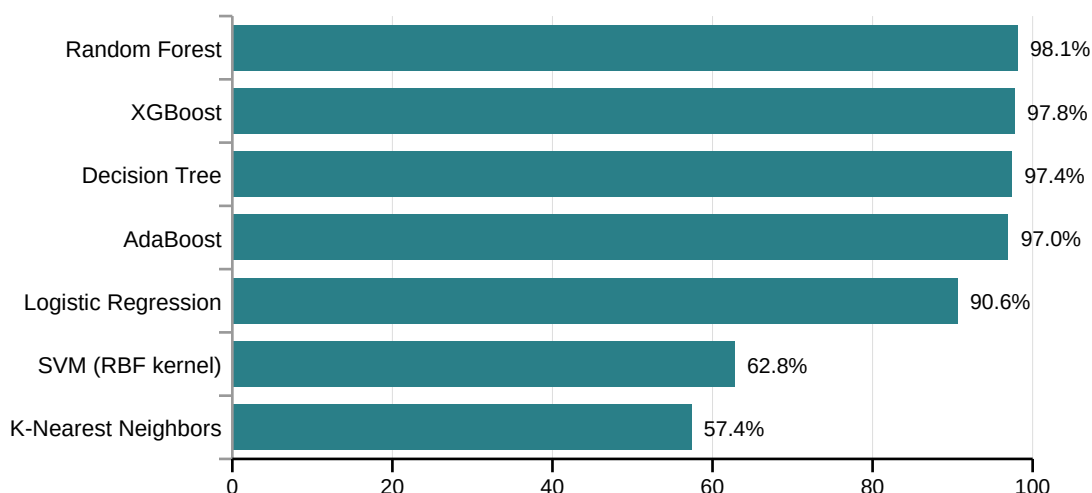
The dataset was split into training (80%) and test (20%) sets using a fixed `random_state` of 42 for reproducibility.

7. Model Building & Comparison

Seven classification algorithms were trained on the same training data and evaluated on the held-out test set using accuracy as the primary metric.

Model	Test Accuracy
Random Forest	98.13%
XGBoost	97.78%
Decision Tree	97.42%
AdaBoost	96.96%
Logistic Regression	90.63%
SVM (RBF kernel)	62.76%
K-Nearest Neighbors	57.38%

Test accuracy by model



Tree-based methods (Random Forest, XGBoost, AdaBoost, Decision Tree) substantially outperform distance-based methods (KNN, SVM) on this dataset, likely because the decision boundary for loan approval is dominated by threshold-like rules on CIBIL score rather than smooth distance relationships in feature space. The SVM and KNN scores are unusually low, however, and should be verified (see Section 13).

8. Hyperparameter Tuning

Hyperparameter tuning was performed on the Random Forest model to identify the optimal configuration automatically and reduce overfitting.

8.1 GridSearchCV

- Best parameters: `max_depth = 15`, `min_samples_split = 2`, `n_estimators = 50`
- Best cross-validated accuracy: 97.80%
- Test accuracy of the tuned model: 97.42%

8.2 RandomizedSearchCV

- Best parameters: `n_estimators = 100`, `min_samples_split = 2`, `min_samples_leaf = 2`, `max_depth = 15`
- Best cross-validated accuracy: 97.75%
- Test accuracy of the tuned model: 97.78%

Configuration	CV Accuracy	Test Accuracy
Random Forest (default)	-	98.13%
GridSearchCV	97.80%	97.42%
RandomizedSearchCV	97.75%	97.78%

Tuning did not raise test accuracy above the default Random Forest (98.13%). Its value here is a more constrained, cross-validated configuration (`max_depth = 15`) that is less prone to overfitting, with cross-validated accuracy close to test accuracy.

9. Model Evaluation

The tuned model was further evaluated using an ROC curve and AUC score to assess its ability to distinguish approved from rejected applications across all classification thresholds.

- **AUC score: 0.998**

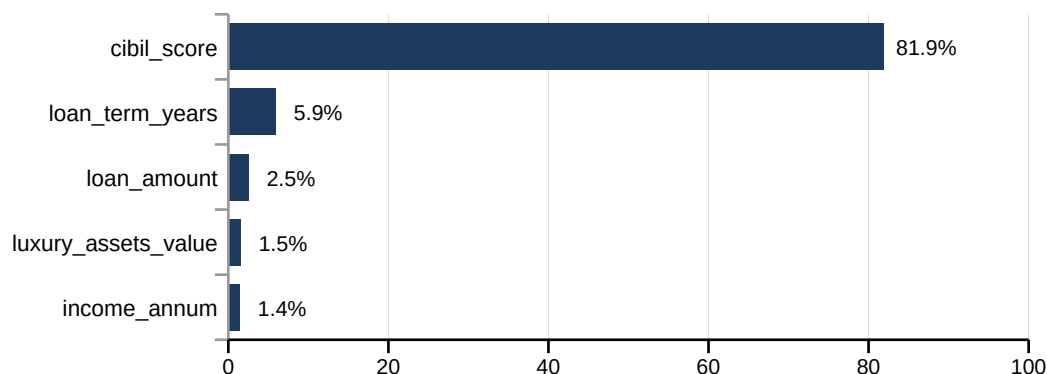
An AUC this close to 1.0 indicates the model separates the two classes almost perfectly, which is consistent with CIBIL score acting as a near-deterministic signal for approval in this dataset.

10. Feature Importance

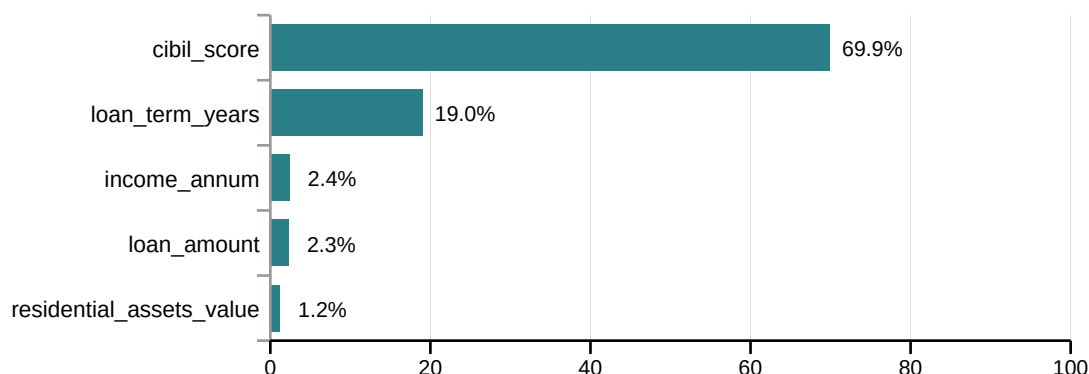
Feature importance was extracted from both the Random Forest and XGBoost models to understand which factors drive the prediction.

Rank	Random Forest	XGBoost
1	cibil_score (81.9%)	cibil_score (69.9%)
2	loan_term_years (5.9%)	loan_term_years (19.0%)
3	loan_amount (2.5%)	income_annum (2.4%)
4	luxury_assets_value (1.5%)	loan_amount (2.3%)
5	income_annum (1.4%)	residential_assets_value (1.2%)

Random Forest: top 5 feature importances



XGBoost: top 5 feature importances



Both models agree that `cibil_score` is overwhelmingly the most important feature, followed by `loan_term_years`. Demographic factors such as education, self-employment status, and number of dependents contribute minimally to the prediction.

11. Key Business Insights & Recommendations

- Give the highest priority to an applicant's CIBIL score when assessing loan approval; it is the single strongest driver of the outcome.
- Factor in loan term, which carries the second-highest weight in both models.
- Ensure the requested loan amount is proportionate to the applicant's income.
- Consider asset values (residential, commercial, luxury, bank balance) as secondary indicators of repayment capacity.
- Applicants who maintain a higher bank balance show a higher likelihood of approval.
- Education and self-employment status have little practical effect on approval and should not drive decisions.

- **Recommended priority order for manual review:** (1) CIBIL score, (2) loan term and loan amount relative to income, (3) asset holdings and bank balance.

12. Model Export

The trained Random Forest model was serialized using joblib (Loan_model.pkl) so it can be loaded directly for inference without retraining, enabling integration into a downstream application or API.

13. Limitations & Validation Checks

The results above are strong, but several items should be verified before this model is used in a real lending workflow.

Recommended checks

- **Additional metrics.** Only accuracy is reported. Add a confusion matrix, precision, recall and F1, especially since the goal is to minimize approvals of high-risk applicants (false positives). Confirm the AUC of 0.998 was computed on the test set.
- **SVM and KNN scores.** SVM at 62.76% is close to a majority-class baseline, which is unusual on scaled data. Check that the scaler was applied consistently to training and test sets.
- **Label encoding.** Scikit-learn's LabelEncoder assigns codes alphabetically (Approved = 0, Rejected = 1). Confirm the mapping matches the Approved = 1 convention used in this report.
- **Data leakage.** Ensure StandardScaler was fit on the training set only, after the split.
- **CIBIL dependence.** The model is essentially learning a CIBIL threshold. It should not be treated as an independent assessment of repayment risk.
- **Fairness and auditability.** Loan decisions require explainability and bias review before deployment.

14. Conclusion & Future Work

The project demonstrates that loan approval outcomes in this dataset can be predicted with very high accuracy, primarily driven by the applicant's CIBIL score. Random Forest and XGBoost are the recommended models for production use given their strong accuracy and interpretability via feature importance.

Future Improvements

- Evaluate class-imbalance handling techniques (e.g., SMOTE) if the approval/rejection ratio shifts in future data.
- Experiment with LightGBM, CatBoost, or stacked ensembles for further gains.
- Add SHAP-based explainability to support individual, auditable loan decisions.
- Deploy the saved model behind a simple API or web app (e.g., Flask/Streamlit) for real-time predictions.